

UNIT 2 :- Introduction to Data Mining

- Basic concepts of data mining
- Applications of Data Mining
- Differences between Data Mining and Data Warehousing
- Data Mining Issues
- Types of Data to be mined
- Stages of the Data Mining Process
- Data Mining Techniques.

Data mining

Data mining is the process of extracting useful information from an accumulation of data, often from a data warehouse or collection of linked data sets. Data mining tools include powerful statistical, mathematical, and analytics capabilities whose primary purpose is to sift through large sets of data to identify trends, patterns, and relationships to support informed decision-making and planning.

Often associated with marketing department inquiries, data mining is seen by many executives as a way to help them better understand demand and to see the effect that changes in products, pricing, or promotion have on sales. But data mining has considerable benefit for other business areas as well. Engineers and designers can analyze the effectiveness of product changes and look for possible causes of product success or failure related to how, when, and where products are used. Service and repair operations can better plan parts inventory and staffing. Professional service organizations can use data mining to identify new opportunities from changing economic trends and demographic shifts.

Data mining becomes more useful and valuable with bigger data sets and with more user experience. Logically, the more data, the more insights and intelligence should be buried there. Also, as users get more familiar with the tools and better understand the database, the more creative they can be with their explorations and analyses.

The Scope of Data Mining

Data mining derives its name from the similarities between searching for valuable business information in a large database — for example, finding linked products in gigabytes of store scanner data — and mining a mountain for a vein of valuable ore. Both processes require either sifting through an immense amount of material, or intelligently probing it to find exactly where the value resides. Given databases of sufficient size and quality, data mining technology can generate new business opportunities by providing these capabilities:

1) Automated prediction of trends and behaviours

Data mining automates the process of finding predictive information in large databases. Questions that traditionally required extensive hands on analysis can now be answered directly from the data — quickly. A typical example of a predictive problem is targeted marketing. Data mining uses data on past promotional mailings to identify the targets most likely to maximize return on investment in future mailings. Other predictive problems include forecasting bankruptcy and other forms of default, and identifying segments of a population likely to respond similarly to given events.

2) Automated discovery of previously unknown patterns.

Data mining tools sweep through databases and identify previously hidden patterns in one step. An example of pattern discovery is the analysis of retail sales data to identify seemingly unrelated products that are often purchased together. Other pattern discovery problems include detecting fraudulent credit card transactions and identifying anomalous data that could represent data entry keying errors.

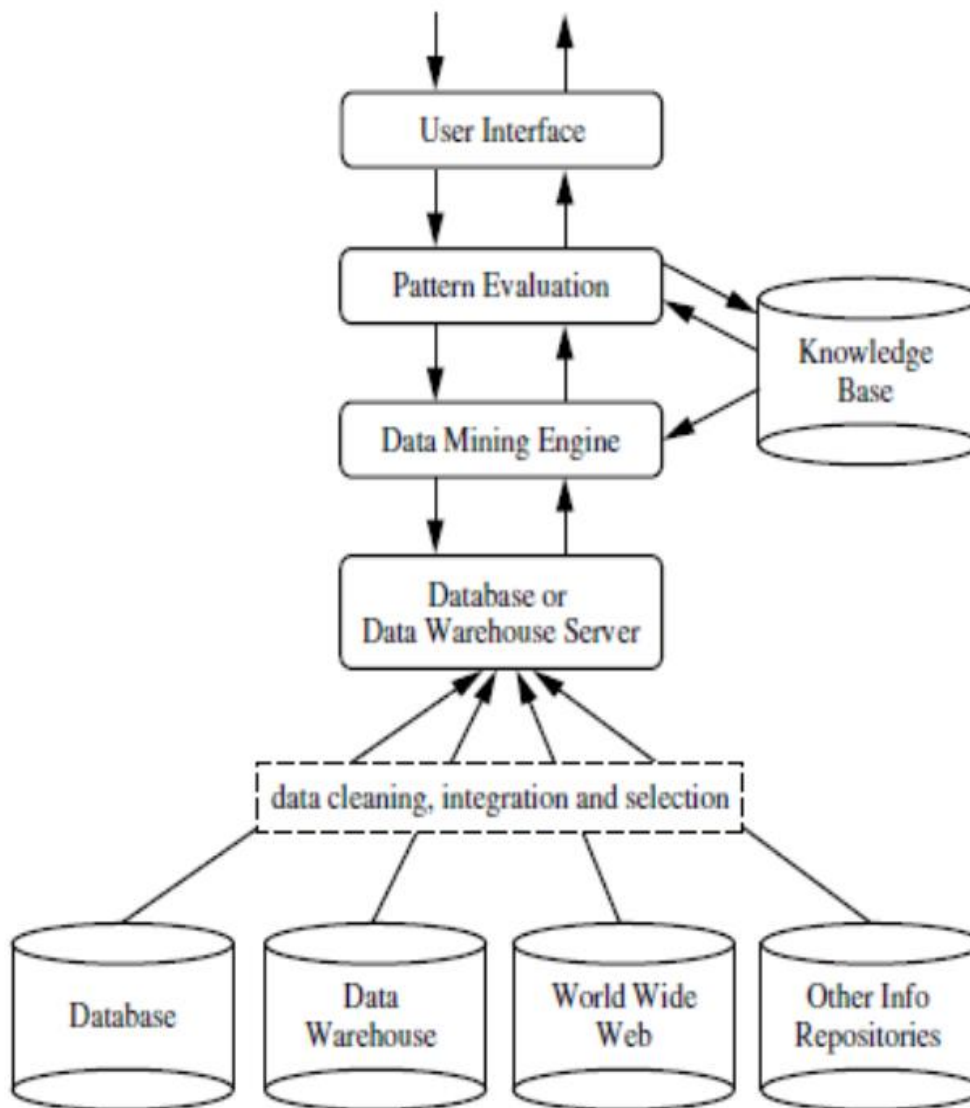
Tasks of Data Mining :

Data mining involves six common classes of tasks:

- 1) **Anomaly detection (Outlier/change/deviation detection) –**
The identification of unusual data records, that might be interesting or data errors that require further investigation.
- 2) **Association rule learning (Dependency modelling) –**
Searches for relationships between variables. For example a supermarket might gather data on customer purchasing habits. Using association rule learning, the supermarket can determine which products are frequently bought together and use this information for marketing purposes. This is sometimes referred to as market basket analysis.
- 3) **Clustering –**
It is the task of discovering groups and structures in the data that are in some way or another "similar", without using known structures in the data.
- 4) **Classification –** is the task of generalizing known structure to apply to new data. For example, an e-mail program might attempt to classify an e-mail as "legitimate" or as "spam".
- 5) **Regression –** attempts to find a function which models the data with the least error. DEPT OF CSE & IT VSSUT, Burla.
- 6) **Summarization –** providing a more compact representation of the data set, including visualization and report generation.

Architecture of Data Mining :

A typical data mining system may have the following major components.



1) **Knowledge Base:**

This is the domain knowledge that is used to guide the search or evaluate the interestingness of resulting patterns. Such knowledge can include concept hierarchies, DEPT OF CSE & IT VSSUT, Burla used to organize attributes or attribute values into different levels of abstraction. Knowledge such as user beliefs, which can be used to assess a pattern's interestingness based on its unexpectedness, may also be included. Other examples of domain knowledge are additional interestingness constraints or thresholds, and metadata (e.g., describing data from multiple heterogeneous sources).

2) Data Mining Engine:

This is essential to the data mining system and ideally consists of a set of functional modules for tasks such as characterization, association and correlation analysis, classification, prediction, cluster analysis, outlier analysis, and evolution analysis.

3) Pattern Evaluation Module:

This component typically employs interestingness measures interact with the data mining modules so as to focus the search toward interesting patterns. It may use interestingness thresholds to filter out discovered patterns. Alternatively, the pattern evaluation module may be integrated with the mining module, depending on the implementation of the data mining method used. For efficient data mining, it is highly recommended to push the evaluation of pattern interestingness as deep as possible into the mining process so as to confine the search to only the interesting patterns.

4) User interface:

This module communicates between users and the data mining system, allowing the user to interact with the system by specifying a data mining query or task, providing information to help focus the search, and performing exploratory data mining based on the intermediate data mining results. In addition, this component allows the user to browse database and data warehouse schemas or data structures, evaluate mined patterns, and visualize the patterns in different forms.

- **Stages of the Data Mining Process:**

Data Mining is a process of discovering various models, summaries, and derived values from a given collection of data.

The general experimental procedure adapted to data-mining problems involves the following steps:

- 1. State the problem and formulate the hypothesis –**

Most data-based modelling studies are performed in a particular application domain. Hence, domain-specific knowledge and experience are usually necessary in order to come up with a meaningful problem statement. Unfortunately, many application studies tend to focus on the data-mining technique at the expense of a clear problem statement. In this step, a modeller usually specifies a set of variables for the unknown dependency and, if possible, a general form of this dependency as an initial hypothesis. There may be several hypotheses formulated for a single problem at this stage. The first step requires the combined expertise of an application domain and a data-mining model. In practice, it usually means a close interaction between the data-mining expert and the application expert. In successful data-mining applications, this cooperation does not stop in the initial phase; it continues during the entire data-mining process.

- 2. Collect the data –**

This step is concerned with how the data are generated and collected. In general, there are two distinct possibilities. The first is when the data-generation process is under the control of an expert (modeler): this approach is known as a designed experiment. The second possibility is when the expert cannot influence the data-generation process: this is known as the observational approach. An observational setting, namely, random data generation, is assumed in most data-mining applications. Typically, the sampling distribution is completely unknown after data are collected, or it is partially and implicitly given in the data-collection procedure. It is very important, however, to understand how data collection affects its theoretical distribution, since such a priori knowledge can be very useful for modelling and, later, for the final interpretation of results. Also, it is important to make sure that the data used for estimating a model and the data used later for testing and applying a model come from the same, unknown, sampling distribution. If this is not the case, the estimated model cannot be successfully used in a final application of the results.

3. Preprocessing the data –

In the observational setting, data are usually "collected" from the existing databases, data warehouses, and data marts. Data preprocessing usually includes at least two common tasks

- a. **Outlier detection (and removal)** – Outliers are unusual data values that are not consistent with most observations. Commonly, outliers result from measurement errors, coding and recording errors, and, sometimes, are natural, abnormal values. Such non representative samples can seriously affect the model produced later. There are two strategies for dealing with outliers:
 - i. Detect and eventually remove outliers as a part of the preprocessing phase, or
 - ii. Develop robust modelling methods that are insensitive to outliers.

- b. **Scaling, encoding, and selecting features –**

Data preprocessing includes several steps such as variable scaling and different types of encoding. For example, one feature with the range $[0, 1]$ and the other with the range $[-100, 1000]$ will not have the same weights in the applied technique; they will also influence the final data-mining results differently. Therefore, it is recommended to scale them and bring both features to the same weight for further analysis. Also, application-specific encoding methods usually achieve dimensionality reduction by providing a smaller number of informative features for subsequent data modeling.

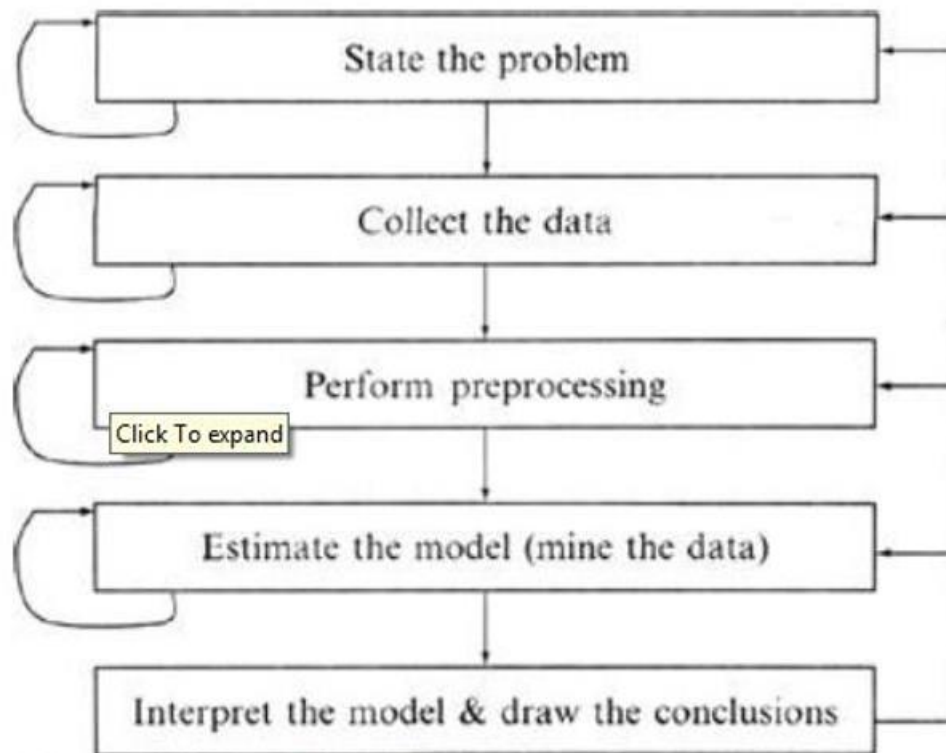
These two classes of preprocessing tasks are only illustrative examples of a large spectrum of preprocessing activities in a data-mining process. Data-preprocessing steps should not be considered completely independent from other data-mining phases. In every iteration of the data-mining process, all activities, together, could define new and improved data sets for subsequent iterations. Generally, a good preprocessing method provides an optimal representation for a data-mining technique by incorporating a priori knowledge in the form of application-specific scaling and encoding.

4. Estimate the model –

The selection and implementation of the appropriate data-mining technique is the main task in this phase. This process is not straightforward; usually, in practice, the implementation is based on several models, and selecting the best one is an additional task.

5. Interpret the model and draw conclusions –

In most cases, data-mining models should help in decision making. Hence, such models need to be interpretable in order to be useful because humans are not likely to base their decisions on complex "black-box" models. Note that the goals of accuracy of the model and accuracy of its interpretation are somewhat contradictory. Usually, simple models are more interpretable, but they are also less accurate. Modern data-mining methods are expected to yield highly accurate results using high dimensional models. The problem of interpreting these models, also very important, is considered a separate task, with specific techniques to validate the results. A user does not want hundreds of pages of numeric results. He does not understand them; he cannot summarize, interpret, and use them for successful decision making.



The Data mining Process

Major Issues In Data Mining:

- **Mining different kinds of knowledge in databases** - The need of different users is not the same. And Different user may be in interested in different kind of knowledge. Therefore it is necessary for data mining to cover broad range of knowledge discovery task.
- **Interactive mining of knowledge at multiple levels of abstraction.**
- The data mining process needs to be interactive because it allows users to focus the search for patterns, providing and refining data mining requests based on returned results.
- **Incorporation of background knowledge.** - To guide discovery process and to express the discovered patterns, the background knowledge can be used. Background knowledge may be used to express the discovered patterns not only in concise terms but at multiple level of abstraction.
- **Data mining query languages and ad hoc data mining.** - Data Mining Query language that allows the user to describe ad hoc mining tasks, should be integrated with a data warehouse query language and optimized for efficient and flexible data mining.
- **Presentation and visualization of data mining results.** - Once the patterns are discovered it needs to be expressed in high level languages, visual representations. This representations should be easily understandable by the users.
- **Handling noisy or incomplete data.** - The data cleaning methods are required that can handle the noise, incomplete objects while mining the data regularities. If data cleaning methods are not there then the accuracy of the discovered patterns will be poor.
- **Pattern evaluation.** - It refers to interestingness of the problem. The patterns discovered should be interesting because either they represent common knowledge or lack novelty.
- **Efficiency and scalability of data mining algorithms.** - In order to effectively extract the information from huge amount of data in databases, data mining algorithm must be efficient and scalable.
- **Parallel, distributed, and incremental mining algorithms.** - The factors such as huge size of databases, wide distribution of data, and complexity of data mining methods motivate the development of parallel and distributed data mining algorithms. These algorithm divide the data into partitions which is further processed parallel. Then the results from the partitions are merged. The incremental algorithms, updates databases without having mine the data again from scratch.

What Are Data Mining Applications?

Every industry collects data from various primary and secondary data sources. The expanse of data collected may be structured or unstructured and requires pre-processing before features can be extracted. Data mining is the process of extracting this data and using advanced computational techniques to derive insights. The results help organisations make effective business decisions, reduce risks and improve performance. For efficient data mining applications, the data sources, data quality and data storage require proper management. Data mining is an effective computational process that has several benefits in varying industries. Some of its applications are:

1. Healthcare

Data mining helps improve the quality of healthcare systems. Predictive analysis helps recommend medicines and evaluate the treatment progress. Similarly, identifying unusual patterns in medical claims, medicine purchases or incoherent prescriptions help track fraudulent practices.

2. Insurance

Data mining helps insurance companies understand customer purchase behaviour and predict the insurance policy they are likely to purchase in the future. They track fraudulent claim practices and strengthen their systems to avoid them. Insurance companies can trace policies that get claimed together and bundle them to provide better services.

3. Market basket analysis

Market basket analysis suggests that if a customer buys a certain quantity of a particular product, they may purchase it again or look for similar products. Understanding this data helps retailers identify the frequency of purchase and manage their stock accordingly. It also helps improve sales and manage customer relations.

4. Financial analysis

Banks have detailed information about their customers, their transactions and loans. Understanding this bulk of data allows banks to classify customers and customise services like loans, credit card spending limits, rewards and provide discounts on purchases. Identifying unusual activity in a transaction helps track fraudulent activities and security breaches.

5. Intrusion detection

Data mining techniques help classify information for intrusion detection systems. The system then generates an alarm on detecting any foreign elements that do not fit the classification rule. This process helps detect security breaches, attacks,

misuse and anomalies. Data mining techniques are crucial for any business and help protect essential information.

6. Energy

Data mining helps track energy consumption patterns and devise systems to increase efficiency. It aids in predicting power consumption in different geographical locations. Elaborate data mining systems provide details on operations patterns too. This insight later helps to optimise operations and invest in equipment that improves production efficiency.

7. Retail and E-commerce

The retail and e-commerce sector collects and tracks customer details, transactions and product sales. It helps them identify customer purchase behaviour, product preferences and seasonal product sales. This data benefits organisations to forecast sales and customise their offerings. Efficiently using past data to make business decisions helps retailers and e-commerce owners reduce risk and increase profitability.

8. Spatial data mining

Data mining facilitates the study of past data, discovery and analysis of spatial and geographical information. It helps identify hotspots and unusual locations, taking the spatial relations between objects into account. Additionally, their latitude, area, perimeter and coordinates, help discover previously unknown but potentially useful information.

9. Biological data analysis

Using complex computational analysis data mining facilitates the study and interpretation of biological datasets. It helps predict protein structure, gene classification and analyses cell mutation. It advances biological studies and improves the healthcare system.

10. Criminal investigation

The main objective of using data mining in criminal investigation is to fasten the rate of solving a crime. Clustering data helps group crime characteristics and devise ways to prevent them. Data from multiple sources is analysed to simplify complex relations between crime and criminal. This helps identify patterns in crime over a time-period or geographical location.

11. Supply chain management

Companies with numerous warehouses requiring the transport of goods and materials use data mining to optimise their process. It helps them analyse demand patterns and plan supply accordingly. Companies also use this data to

make their distribution channels more efficient and improve coordination with other warehouses and distributors.

12. Media

Media channels like radio, television and over-the-top (OTT) platforms keep track of their audience to understand consumption patterns. Using this information, media providers make content recommendations, change programme schedules and produce content of the preferred genre. Data mining helps media providers improve the viewer experience.

13. Advertising and marketing

With the advent of digital marketing and data mining technologies, marketers refine their strategies for better engagement and track the live results of their campaigns. Advertisers also use this data to profile users and show them content or product that might interest them. Data mining is widely used in digital marketing to improve targeting and user experience.

14. Education

Data mining is used in education to learn student productivity and development. It helps understand how a student is performing, predict their future scores, identify relevant placement opportunities and track teacher performance. Data mining may help derive associations between the teaching methodologies and student performance and identify areas of improvement.

15. Business

The timely use of analysis derived from data mining can help businesses make informed decisions. Data mining helps understand customer purchase patterns, their behaviours and understand the products or services they are likely to buy. Segmenting customers into different datasets helps customise sales and marketing strategies that reduce risk and increase profitability.

- **Data Mining Techniques**

Data scientists and analysts widely use different data mining techniques to find patterns that help businesses make informed decisions. Some common techniques include:

1. Classification

Classification is a data mining technique based on machine learning. It involves analysing data attributes to categorise data into classes. Data analysts apply algorithms to classify and categorise data, often using this technique to develop software that classifies items into different groups. For example, a data analyst might classify the "Financial background of customers seeking a loan" into "low," "medium," and "high" credit risks. With details like salary and credit score, software can be trained to predict the likelihood of a customer defaulting on loan repayments.

2. Clustering

Clustering is an important data mining technique as it creates meaningful clusters of data objects that share the same characteristics. It means that in a particular cluster, all objects are similar to one another. The clustering technique helps data analysts discover groups and clusters. The degree of association with objects within the same cluster is high. As a result, this data mining method finds its use in customer profiling. Also, data mining applications like information retrieval, spatial database application, scientific data exploration, web analysis and medical diagnostics widely use clustering techniques to cluster similar data objects.

3. Tracking patterns

Tracking patterns is a fundamental technique of data mining. A data analyst identifies and monitors patterns or trends in data to make data-driven inferences that can affect the company's bottom line. Once a company identifies a particular trend or pattern, it can capitalise and build on the trend. For example, if an analyst uncovers that the sale of a particular product is high among teenagers, they can create similar products for them.

4. Association

The association or relation technique relates to statistics. Using this data mining method, data scientists and analysts can find links between two or more data attributes. The association technique helps find events or data attributes that have a high correlation with another event or attribute. It also uncovers the hidden patterns in your data set. For example, you might notice that when a customer avails of a particular service, they often buy a product related to the service.

5. Outlier analysis

Data scientists try to find anomalies or outliers in a data set using the outlier analysis technique. It is often the first step in several data mining applications. An outlier is a

data point that provides information and displays abnormal or different behaviour from other data points. Data analysts and scientists can use the univariate or multivariate method to find potential outliers during the data mining process. Financial applications, credit and debit card frauds and network interruption identification widely use the outlier detection technique.

6. Prediction

Prediction is a valuable and important data mining technique that discovers the relationship between dependent and independent attributes and between independent attributes alone. This technique scrutinises past and historical trends or patterns in the right sequence to predict a future event. Using predictive analysis, a data scientist can understand future trends that help in making key business decisions. For example, you might review a customer's credit score and financial history to understand credit risk in the future.

7. Regression

Unlike prediction, regression is a data mining method that discovers the relation between the attributes in a given data set. For example, data analysts can use regression to project the price of a product based on other factors, such as availability, demand and inflation. It helps a data scientist uncover the relationship between two or more data attributes. Data modelling and forecasting are two major applications of this technique.

8. Decision trees

A decision tree is a supervised learning algorithm that allows data mining and data science professionals to analyse the data effectively. Decision trees help these professionals understand how the input affects the output, more like an if-then analysis. The root of a decision tree can be a question or a condition with multiple answers. Each answer leads to another set of questions and conditions. This helps in analysing data and allows data mining professionals to come to logical inferences.

9. Sequential patterns

This data mining technique identifies sequences of events or patterns in transaction data over time. For example, historical sales data may show that customers who buy a camera often purchase a printer within 30 days. Data scientists use this insight to optimise product placement and promotions in retail. Sequential pattern analysis also applies to weather prediction, web access analysis, network security and production processes.

10. Neural networks

A neural network or artificial neural network is a mathematical model inspired by biological neural networks. Artificial intelligence and deep learning widely use this concept for data mining. Analysts can use neural for customer research data, sales forecasting, time-series prediction and anomaly detection in a data set.

11. Long-term memory processing

Data processing and analysis are immediate, with results changing whenever new data comes in. Instead of conducting analysis again with new data, analysts prefer the long-term memory processing method. Long-term memory processing is the ability to conduct analysis over an extended time. Using this technique, an analyst identifies patterns and trends in the data, which otherwise would be difficult to detect.

12. Data visualisation

Data visualisation represents data and related information in a visual form that is easy to understand and comprehend. As this technique is dynamic, it allows analysts to stream data in real-time. Often, data analysts use dashboards, charts and graphs to uncover useful business information.

13. Data warehouse

A data warehouse collects data from different sources in a company to provide meaningful insights from the data. These warehouses provide analysts with generalised and consolidated data in a multidimensional view. Along with the multidimensional views, a data warehouse provides online analytical processing tools (OLAP) for effective data analysis.

What kind of data can be mined ?

1. Structured data
2. Unstructured data
3. Semi structured data
4. Time series data
5. Spatial data
6. Web data
7. Biological data

1. Structured Data

- Structured data is highly organized and easily searchable data, typically stored
- in fixed formats such as tables or spreadsheets.
- It is often found in relational databases, where data is arranged in rows and columns, making it straightforward to enter, query, and analyze.
- Examples include sales transactions, customer records, and financial data.
- It's like having all your files neatly organized in folders and cabinets, with clear labels on each one.
- Fixed Schema: This schema specifies things like data types (numbers, text, dates), field names (customer name, product ID).

Benefits of Structured Data for Data Mining

- a) **Efficiency:** The well-defined structure allows for efficient storage, retrieval, and manipulation of data using data mining tools and algorithms.
- b) **Scalability:** Structured data formats are easily scalable, meaning you can add more data to existing structures without compromising its integrity.

2. Unstructured data

- Unstructured data refers to information that lacks a predefined format or organization,
- making it more challenging to analyze directly using traditional data mining techniques.
- It's like having a box full of random documents, photos, and recordings – the information is valuable, but it takes extra effort to understand and organize it.
- Variable Format: Unlike structured data with a fixed schema, unstructured data can come in various formats.
- Social Media Posts: Textual content, images, and videos shared on social media platforms.

Importance of Unstructured Data in Data Mining

- a) **Rich Information Source:** Unstructured data often contains valuable insights and hidden patterns that structured data might miss.
- b) **Real-World Representation:** Unstructured data often reflects real-world scenarios where information isn't neatly organized.

3. Semi Structured Data

- Semi-structured data falls somewhere between the well-defined world of structured data and the free-flowing realm of unstructured data.
- It has some internal organization, but it doesn't conform to a strict, pre-defined schema like a relational database table.
- This structure can be achieved through tags, markup languages (like XML or JSON),
- No Fixed Schema: Unlike structured data, there's no single, pre-defined schema dictating the format and meaning of each data element.

Examples: Common examples of semi-structured data in data mining include:

- **Emails:** While emails have a basic structure (sender, recipient, subject, body), the content within the body can be unstructured text, images, or attachments.
- **Web Pages:** Web pages are built with HTML tags that provide some structure, but the content within those tags (text, images, videos) can vary greatly.

4. Time Series Data

- Time series data is a specific type of information that captures observations over time.
- It's like a movie, where each frame represents a data point captured at a specific moment.
- Unlike other data formats that might show various characteristics of individual subjects, time series data focuses on how something changes over time.
- **Sequential Data Points:** Time series data consists of a sequence of data points, each associated with a specific timestamp.
- These timestamps could be seconds, minutes, hours, days, months, years, etc.

Examples

- **Financial Markets:** Stock prices, exchange rates, or trading volumes recorded at regular intervals (e.g., every minute, hour, or day).
- **Sensor Data:** Measurements collected by sensors at regular intervals, such as temperature readings from a machine, air quality, etc.

Importance of Time Series Data

- **Predictive Power:** By analyzing past trends and patterns, time series data allows for making informed predictions about future behavior.

5. Spatial Data

- Spatial data refers to information that explicitly references geographic locations.
- It's like a map where each data point has an address, not just a name.
- Unlike other data types that focus on properties or characteristics, spatial data emphasizes the "where" aspect of information.
- Data Types: Spatial data can be combined with other data types to provide a richer picture.

Applications of Spatial Data

- a) Location-based Services:** Used in apps like navigation systems, ride-hailing services, and restaurant finders.
- b) Urban Planning:** Analyzing population density, traffic patterns, and land use to make informed decisions about city development.

Importance of Spatial Data

- **Integration with Other Data:** Spatial data can be integrated with other data types to create a holistic view of a situation or phenomenon.

6. Web Data

- Web data refers to the vast amount of information available on the World Wide Web.
- This data encompasses a diverse range of content, structures, and formats, making it a rich source for data mining activities.

Types of Web Data

▪ Web Content

This includes the textual information displayed on web pages, such as articles, news stories, product descriptions, etc.

Data mining techniques can be used to extract keywords, identify trends, and understand user behaviour based on this content.

▪ Web Structure

This refers to the way web pages are linked together.

Data mining techniques can analyze these link structures to understand relationships between websites and identify influential domains.

▪ Web Usage Data

This encompasses information about how users interact with websites.

Examples include clickstream data (pages visited), search queries entered, and time spent on specific pages.

Benefits of Web Mining

- **Rich Source of Information:** Web data offers a vast and diverse source of information that can be used for various purposes.
- **Understanding User Behaviour:** By analyzing web usage data, businesses can gain valuable insights into how users interact with their websites and online platforms.

7. Biological Data

- Biological data serves as the foundation for a vast and exciting realm within data mining.
- It encompasses a wide range of information about living organisms, offering valuable insights into their genes, proteins, functions, and interactions.
- Biological data comes in various forms, each offering a unique piece of the puzzle when it comes to understanding life:
- **Genomic data:** This refers to the blueprint of life, containing the DNA sequences, gene expression levels, etc
- **Proteomic data:** This focuses on proteins, the workhorses of cells. It includes information about protein structures, etc.
- **Environmental data:** This considers the external factors influencing organisms, including temperature, and interactions with other living things.